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Professional Summary

Software Engineer with a background in Computer Engineering (PUC Chile) and experience in developing **products with AI components, scalable platforms, distributed systems, and large-scale analytics**. I have worked building full stack systems that integrate **NLP, PyTorch, OpenAI API**, and efficient data processing, together with clean architectures based on **DDD** and **hexagonal patterns**. I am interested in designing robust software for complex problems in **applied AI**, automation, optimization, and backend/product engineering.

Experience

Full Stack Developer

Sep. 2025 – Present

Dily - Automotive Fintech Platform

Santiago, Chile

- End-to-end development of fintech/lending platforms applying **DDD** and **hexagonal architecture** to maintain clean, testable, and evolvable domain logic.
- Backend in **TypeScript/Node.js** with **Express, Inversify, Mikro-ORM/Knex, MySQL/Postgres**, and **AWS** (S3/SQS/Lambda/SES), integrating **Sentry, Twilio**, and **Slack**.
- Frontend with **Angular** in an **Nx** monorepo, implementing **micro-frontends, RxJS/NGRX**, and testing with **Jest/Mocha/Chai/Supertest** and **Cypress/Vitest/Karma**.

Lead Engineer, Alerting Platform Migration & Operations Modernization

2026 – Present

GridWorks - Freelance

Remote

- Led the end-to-end rebuild of an industrial alerting platform in production, replacing a legacy stack based on **Base44/Node-RED** with a modular multi-service architecture.
- Implemented alert lifecycle logic, configuration tooling, escalation chains, durable ingestion, export/import, domain observability, and secure controls for WhatsApp workflows.
- Executed a gradual migration while preserving functional parity with the legacy system and enabling safe cutovers without disrupting production users.

Data Science Researcher

Jul. 2023 – Present

Pontificia Universidad Católica de Chile

Santiago, Chile

- Lead **Politiktok** (Fondecyt), building an analytics platform that processes datasets of 100,000+ records using **Python, PostgreSQL**, and optimization techniques, achieving 10x performance improvements.
- Designed a scalable full stack architecture for large-scale social media behavior analysis, integrating **NLP, PyTorch**, data visualization, and RESTful APIs.

AI Systems Developer

Jan. 2024 – Mar. 2024

Drone Response - Univ. of Notre Dame

Notre Dame, Indiana, USA

- Developed a **Smart Mission Planner** for drone fleet coordination by implementing route optimization algorithms in **Python**.
- Integrated **OpenAI API, MQTT**, and backend services in **Python/Java Spring** to coordinate multiple autonomous agents and enable intelligent decision-making.

Software Engineer (Full Stack)

Dec. 2024 – Jul. 2025

Flair - StartupChile Growth Winner

Santiago, Chile

- Worked on an energy optimization platform with a microservices architecture using **Python** and **Vue.js/React**, deployed on **AWS** with **Docker** containers.
- Integrated **InfluxDB** and **DynamoDB** for real-time data, optimizing queries and designing efficient architectures for high-frequency data flows.

Advanced Teaching Assistant & Technical Mentor

Mar. 2023 – Present

Pontificia Universidad Católica de Chile

Santiago, Chile

- Advanced teaching assistant in **Operating Systems, Networks, High Performance Computing**, and **Extended Reality**.
- Have completed **more than 14 teaching assistantships** in total, supporting more than 100 students in systems programming, visualization, and distributed architectures.

Technical Instructor - Python & AI Applications

2024 – Present

Escuela Militar de Chile

Santiago, Chile

- Designed and taught courses on **Python** and **AI applications**, accumulating **3 years / 6 semesters** of teaching experience in this context.

Geoscience Researcher

2023 – 2024

Pontificia Universidad Católica de Chile — Seismology

Santiago, Chile

- Built an optimized geological simulation pipeline using **Python** and **high-performance computing** techniques, with performance improvements of up to 15x.

Relevant Projects

- ATC Smart AI Based eCommerce Platform System | AI, Headless Commerce, B2B/B2CFeb. 2026 – Present
 - Headless eCommerce platform focused on sales, catalog, checkout, salesforce enablement, insights, and business process automation.
 - Designed AI capabilities for operational support, automation, and analytics applied to commercial business workflows.
- GridWorks Alerting Platform Migration | Distributed Systems, WhatsApp Workflows, Railway2026 – Present
 - Re-architecting and productionizing a multi-service industrial alerting platform, covering ingestion, rules, escalation, administrative tooling, and safe migration from a legacy system in production.
- Politiktok - Big Data Analytics Platform | Python, PostgreSQL, NLP, PyTorch2023 – 2025
 - Large-scale analytics platform processing 100,000+ records with parallel pipelines, complex query optimization, and visualization of social media behavior.
- Smart Mission Planner - Autonomous Drones | Python, OpenAI API, MQTT, Java Spring2024
 - Intelligent drone route planning system with event-driven architecture and real-time distributed coordination.
- Tectonic Plate Simulation Engine | Python, High Performance Computing2023 – 2024
 - Tectonic plate simulation engine optimized with performance improvements of up to 15x using parallelization and algorithmic optimization.

Education

- Pontificia Universidad Católica de ChileMarch 2020 – Jul. 2025

Computer Engineering, Graduated with distinction. Software Engineering and Data ScienceSantiago, Chile

International Experience and Certifications

- KHIPU 2025 - Latin American Meeting in Artificial Intelligence: Participation in one of the most relevant Artificial Intelligence and Machine Learning events in Latin America, held in Santiago, focused on advanced training, regional community building, and exchange with researchers and industry.
- UC Bridge Program - University of Notre Dame: Selected among 200+ candidates for an international research program in projects funded by NASA/NSF.
- Advanced English Certification (TOEIC 905): Proven ability to collaborate and present results in international contexts.

Technical Presentations and Publications

- Seismological Society of America Annual Meeting 2023April 2023

Research Presentation - San Juan, Puerto Rico

 - Presented research on high-performance computing systems developed with Python for large-scale simulations.
- TikTok: A Postdigital Mode of Inquiry for Tracing Affects Among Chilean Youth2025

Confirmed publication, currently pending

 - Work derived from the Politiktok project, focused on tracing affects, youth behavior, and TikTok content analysis through data pipelines, computational analysis, and the study of social media interactions.